

MATH 250 HANDOUT 1 - LOGIC

1. Which of these are **statements**? (I.e., for which of these sentences is ‘true or false’ meaningful?)

- (1) (Statement) Today it is raining.
- (2) (not) What is your name?
- (3) (S) Every student in this class is a math major.
- (4) (S) $2 + 2 = 5$.
- (5) (open statement, but not a statement) $x + 1 > 0$.
- (6) (False statement) $\forall x \in \mathbf{R}, x + 1 > 0$.
- (7) (To disprove a for all statement, just give an example where it is false)
- (8) (true statement) $\exists x \in \mathbf{R}$ such that $x + 1 > 0$.
- (9) to prove an "exists" statement, just give an example where it is true.
- (10) $x^2 + 1 > 0$.
- (11) If it is raining, then I will wear my raincoat.
- (12) Whenever it is raining, I will wear my raincoat.
- (13) it is raining $\Rightarrow$ I will wear my raincoat
- (14) If P then Q ; $P \Rightarrow Q$; Whenever P is true, Q is also true.
- (15) Give me that.
- (16) (not) This sentence is false.
- (17) If x is a real number, then $x^2 > 0$.

2. Which of these are true?

- (1) (F) Every student in this class is a math major and a human being. (False because "Every student in this class is a math major" is false)
- (2) (T or F) Every student in this class is a math major or a human being.
- (3) (T or F) $2 + 2 = 5$ or $1 > 0$.
- (4) (T or F) If x is a real number, then $x^2 \geq 0$.
- (5) (T or F) If x is a complex number, then $x^2 \geq 0$.

3. Write the negations of the following. (The negation of a statement is the statement with the opposite truth value.)

- (1) $2 + 2 = 5$; $2 + 2 \neq 5$
- (2) $1 > 0$; $1 \leq 0$
- (3) $2 + 2 = 5$ or $1 > 0$; $2 + 2 \neq 5$ and $1 \leq 0$.
- (4) $P \wedge Q$ (and)
- (5) $P \vee Q$ (or)
- (6) $\neg$ means negation
- (7) $P \vee Q$ (or); $\neg(P \vee Q) = \neg P \wedge \neg Q$
- (8) Every student in this class is a math major.
- (9) There is at least one student in this class who is not a math major.
- (10) Every student in this class is a math major or a human being.

- (11) $\neg(\forall x, P(x)) = \exists x$ such that $\neg P(x)$
- (12) If x is a real number, then $x^2 > 0$. ($x = 0$ is a counterexample)
- (13) $\neg(P \Rightarrow Q) = P \wedge \neg Q$

4. Prove the following using truth tables.

- (1) $P \wedge (Q \vee R) = (P \wedge Q) \vee (P \wedge R)$,
- (2) $(P \vee Q) \vee R = P \vee (Q \vee R)$. (We thus write $P \vee Q \vee R$ for both.)
- (3) $\neg(P \vee Q) = \neg P \wedge \neg Q$,
- (4) $\neg(P \wedge Q) =$ (make a guess similar to problem ??),
- (5) $\neg(\neg P) = P$.